

Design overview

Frame stiffness improvement

The original frame design was designed to be manufactured out of carbon fibre which is solid and has a Young's modulus 20-30x higher than PLA plastic, which explains why the 3D printed frame shows significant flexure with applied load.

Bending stiffness is governed by the flexural rigidity EI , where E is Young's modulus and I is the **second moment of area**. From $M = EI\kappa$, a cantilever arm under tip load F deflects by:

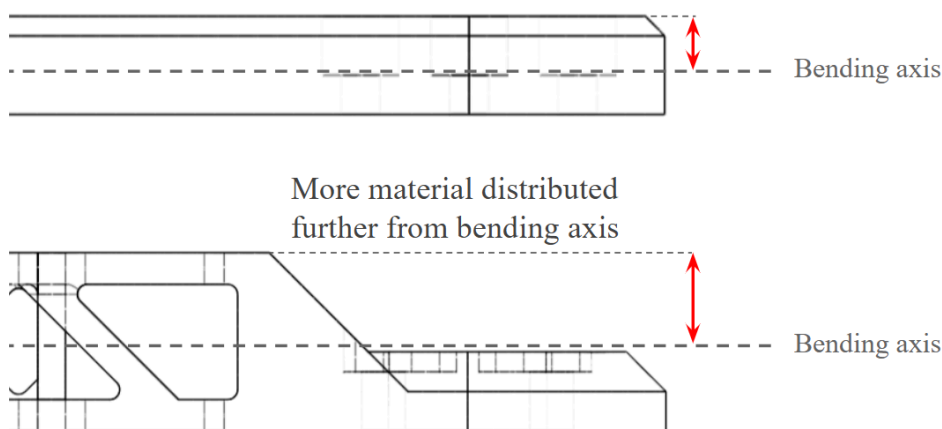
$$\delta = \frac{FL^3}{3EI}$$

Hence, deflection is inversely proportional to I . For a rectangular section, the second moment of area is given by:

$$I = \frac{bt^3}{12}$$

Where b is the width and t is the thickness in the bending direction. From the equation, we see that the bending stiffness scales the most with thickness t .

Finally, vertical triangle cutouts can be used to minimize material and weight as the bending load is borne mainly by the fibres on the extreme ends in tension/compression. Material close to the bending axis contributes exponentially less to the stiffness and can be removed to reduce weight.

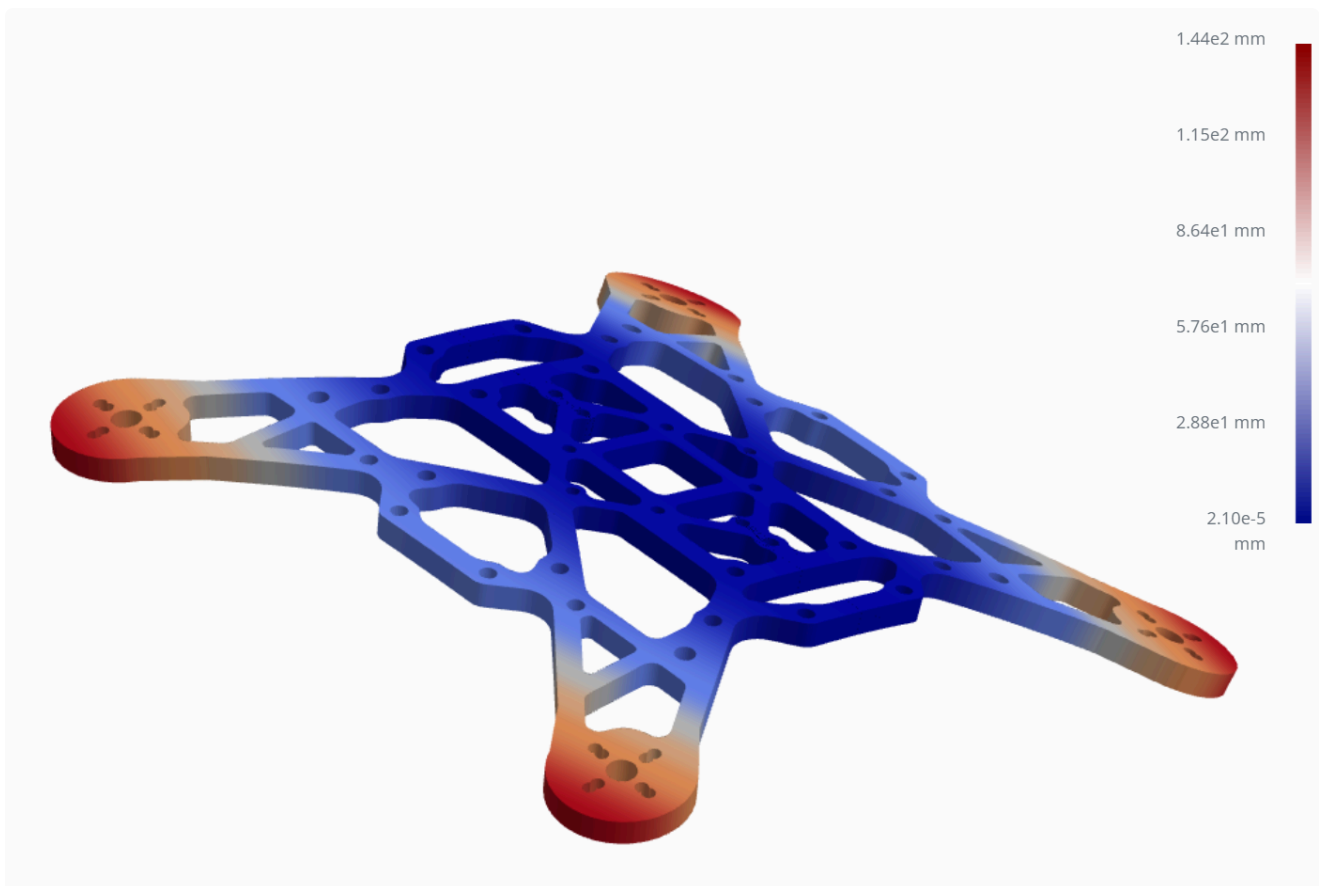


Analysis

A simple bending simulation was performed using [an online mesh simulator](#). The model is fixed about the center and opposing bending moments were applied to either sides. The vertical deflection was recorded in millimeters.

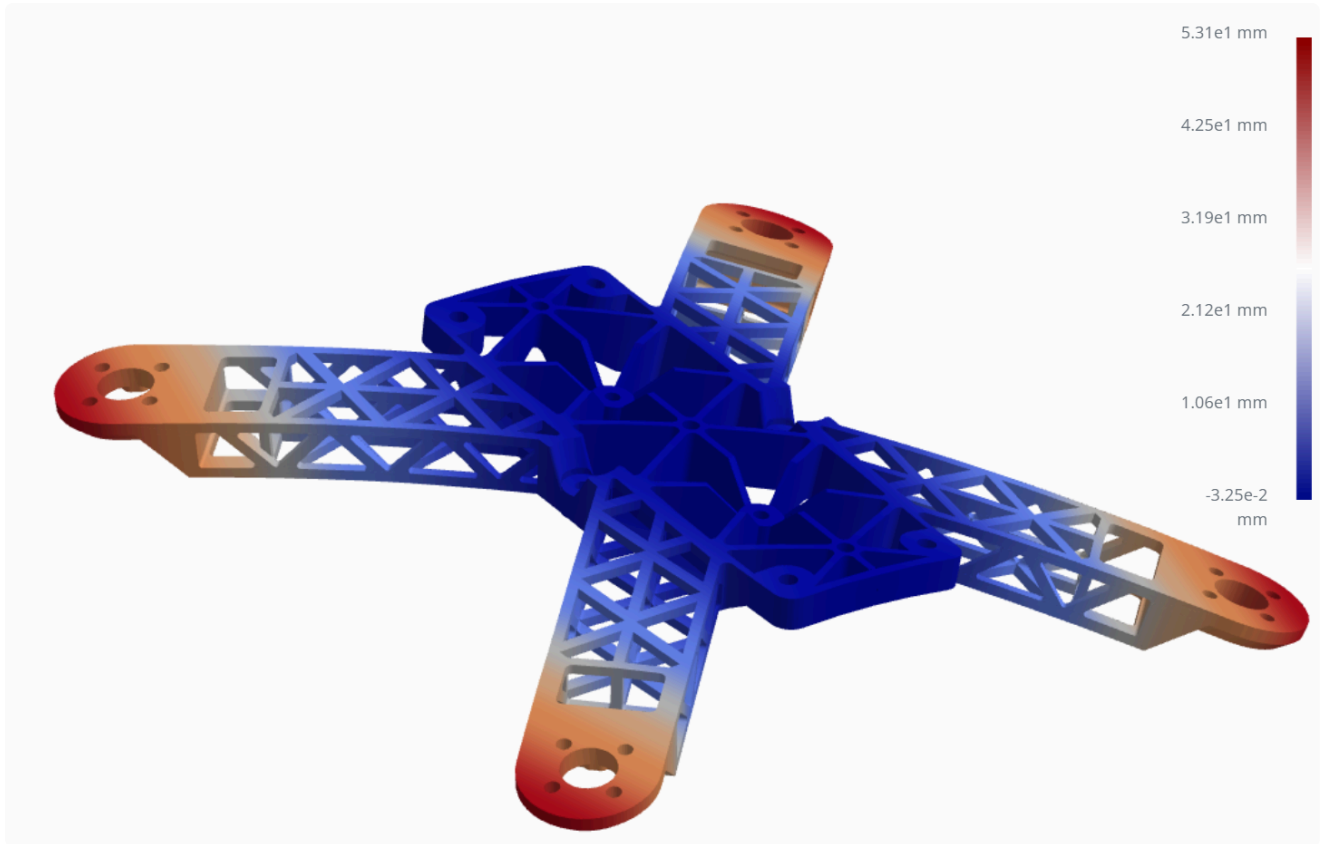
Parameter	Value
Bending moment	30N.m
Material	PLA

Original frame



Max vertical deflection: 14.4mm

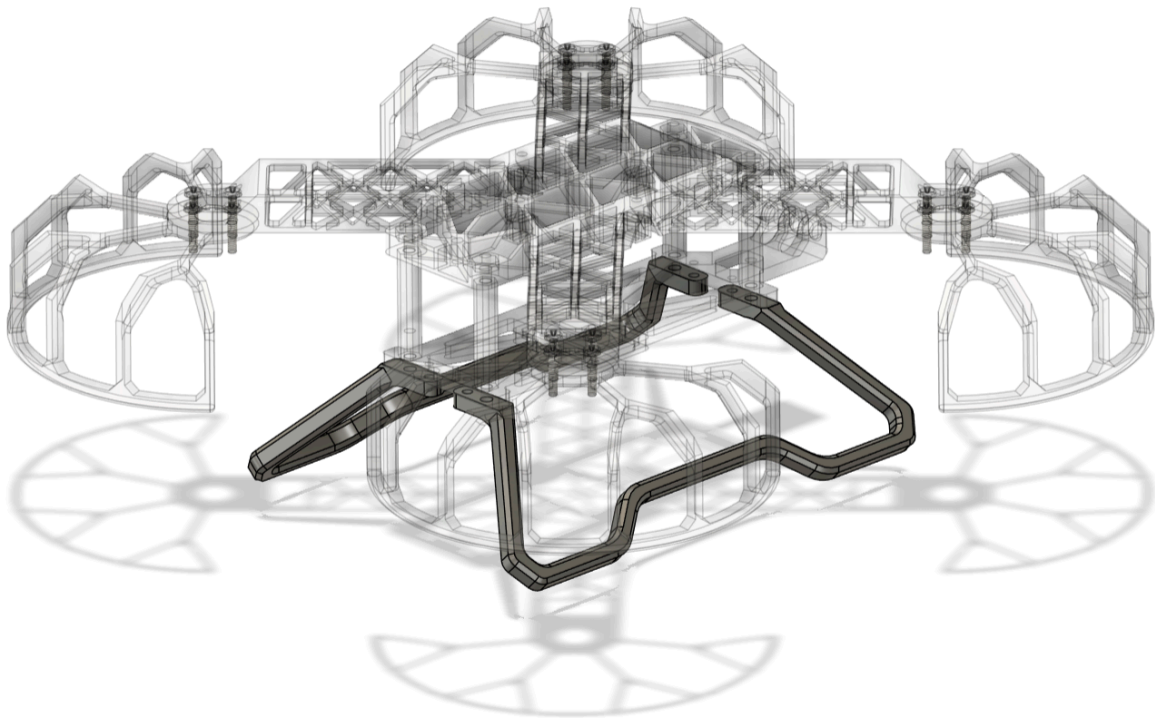
New frame



Max vertical deflection: 5.31mm

The new frame is much stiffer and deflects $\approx 63\%$ less for the same bending moment applied.

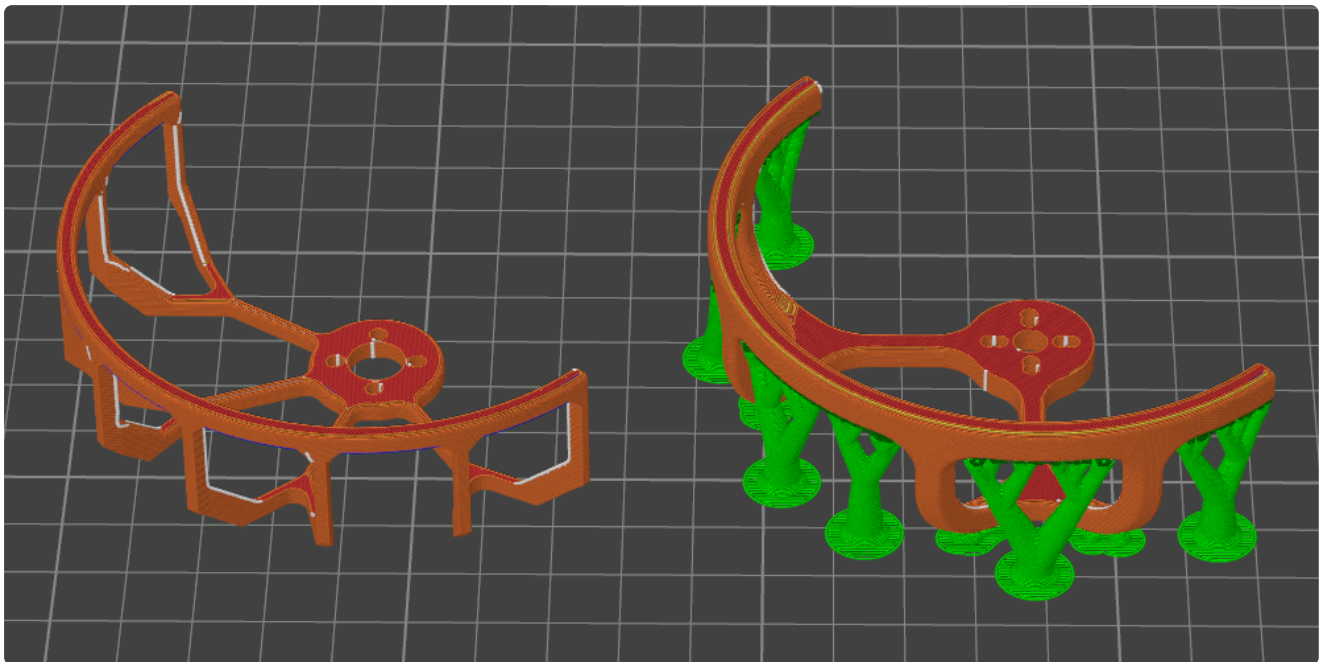
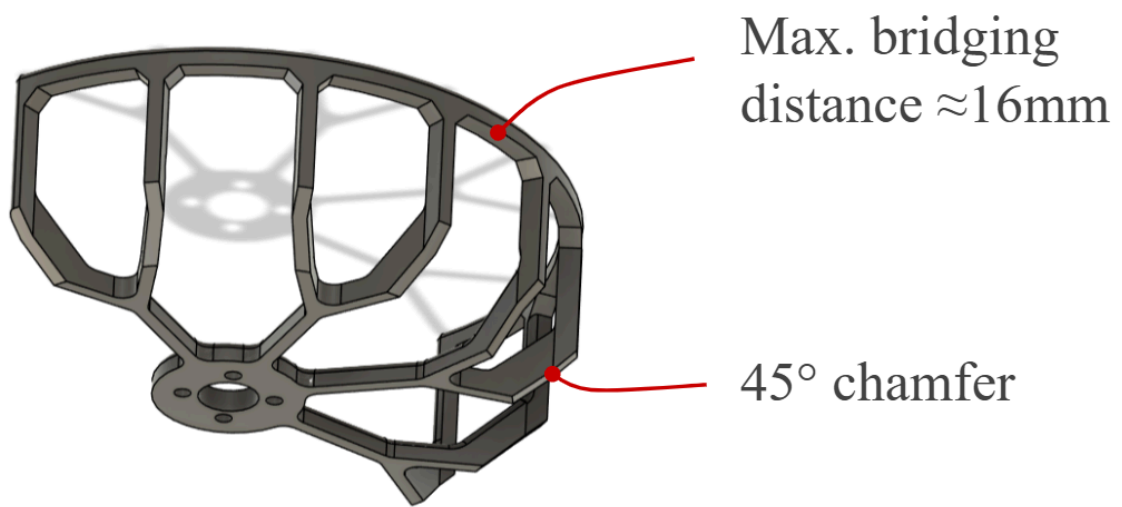
Landing gear



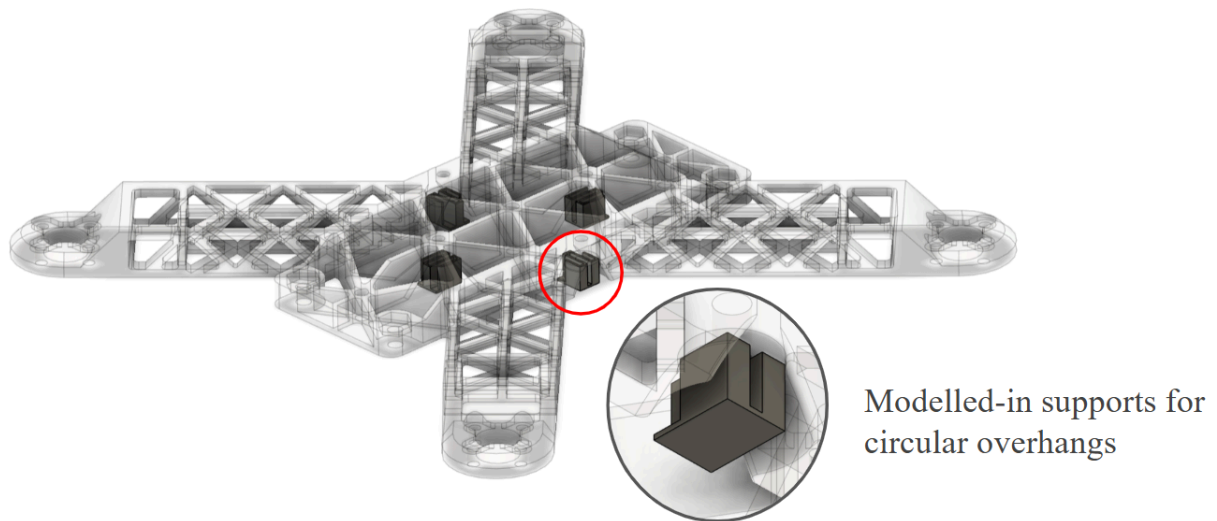
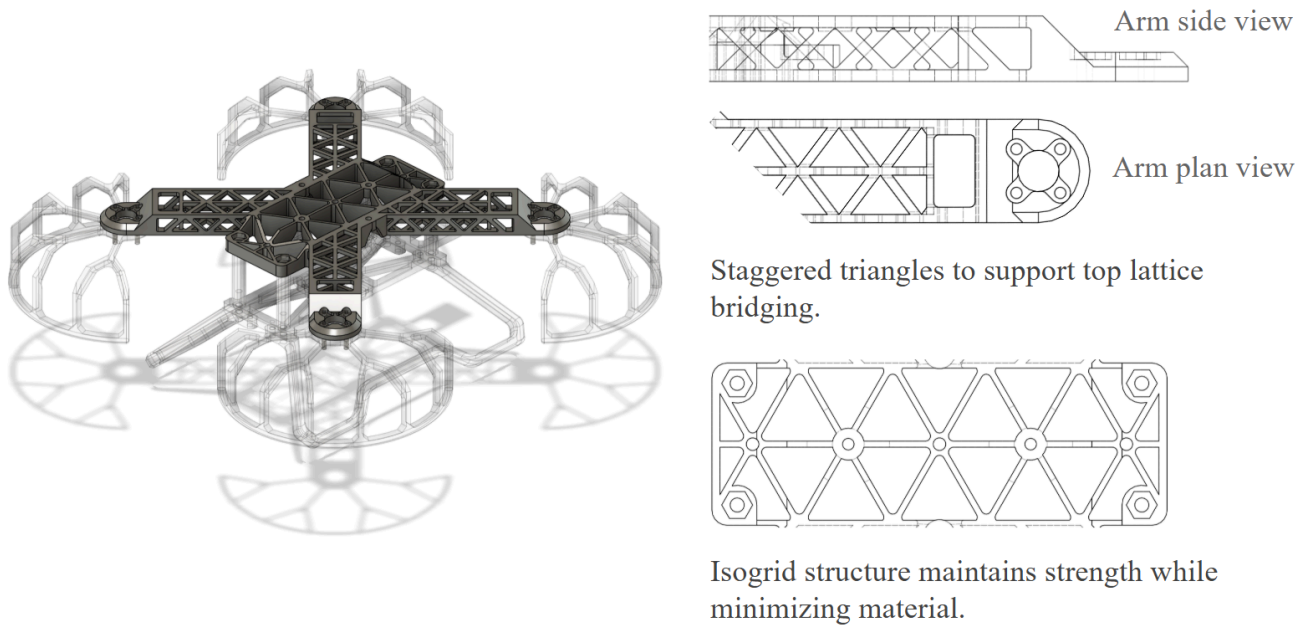
A set of easily replaceable landing feet is designed to splay outwards and absorb landing impact, replacing the nylon spacers in the original design which often came loose on impact.

3D printing optimizations

Common FDM 3D printers can comfortably bridge gaps of $\approx 20mm$ and print overhangs $< 55^\circ$ without droop when printing PLA. All components are designed to be printed without supports by exploiting these features.



New design (left) vs old (right): all supporting material eliminated by controlling bridging distance and overhang angle, saving time and material.



Circular overhangs still require supports since the printer will try to print circles in air instead of bridging the gap. In such cases, supporting structures are directly modelled into the design.

Print time & weight comparison

The new frame design takes the same amount of time (≈ 3.5 hrs) to print as the original, and uses less filament since no supports are required. The total weight of the printed frame components is also similar to the original design at $\approx 50g$.

All frame components fit on a single standard $256 \times 256mm$ print bed.

The final drone assembly with all electrical components fitted (including battery) is around $224g$.

